

Music-Gen v4 closure campaign — cycles 4-6

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Music-Gen v4 closure campaign — cycles 4-6

Abstract

Cycles 4-6 continue the Chicken Grease bass sound-matching sub-milestone from where cycles 1-3 left it. Cycle 4 fires the pre-registered decision protocol on the completed 216-cell stage-2b leaderboard and returns `STILL_INDETERMINATE`: the composite and VGGish objectives disagree — top-1 by composite is program 33 (Electric Bass Finger) with embedding-cosine 0.204, top-1 by embedding-cosine is program 19 (Church Organ) at 0.495 — with neither reaching the 0.60 `CONFIRMED` threshold and neither falling under the 0.40 `RULED_OUT` threshold. In the process of emitting a second pinned profile (`bass_v2.json`, program 33 by composite) and its replay proof, cycle 4 surfaces a `CRITICAL` defect: the sf2 replay dispatch produces byte-identical audio (832868d0...) for two profiles that differ only in `identity.program`. The old and new pinned profiles collide because the replay path silently discards the `program-select` payload.

Cycle 5 formalises the alternate render family. It writes the `family2_stem_sampled` design spec, lands a spike script (`family2_stem_sampled_spike.py`, SHA 000c3ef68042f2da...6329e80) that exercises the intended lever set — single-slice pitch shift + lightweight ADSR + LUFS-I loudness normalisation at -18 LUFS — and escalates the replay defect to the manager with `CRITICAL` severity.

Cycle 6 runs two tracks in a single sequential worker session. Track one replaces the offending `_ = setup # documentation line at replay.py:85` with a real mido-based MIDI rewrite that strips every existing `program_change` and injects a fresh `program_change(channel=0, program=profile.identity.program, time=0)` at tick 0 of the first note-carrying track; three regression tests pass, and the refreshed sf2 replay proofs now diverge as required — `bass` moves to c69775040c...4ff019c while `bass_v2` stays at 832868d0...3aeac5 (expected, because `bass.mid` already embeds `program_change 33`, so the rewrite is materially a no-op for that one profile). Cross-proof SHAs differ, `REPLAY_FIX_LANDS`. Track two promotes the cycle-5 spike into a

shipped `family2_stem_sampled_builder.py`, emits `bass_family2_v1.json` (`profile_id 1f3c104a-...6ee`, `render_family = stem_sampled_v1`), its replay proof (byte-identical $\times 2$ at `9b4647cef61fe9d6...523276`), and the family verdict `FAMILY2_RULED_OUT` at `embedding_cos_vggish = 0.0896`, roughly 0.405 units below the sf2 top-1.

Both frozen render families are now exhausted for Chicken Grease bass without any CONFIRMED profile. The M-V4-SHOWCASE-1 milestone is blocked until the operator picks between accepting the sf2 STILL_INDETERMINATE top-1 as a policy call, refusing showcase until a further family lands, or overriding the pre-registered 0.60 threshold. The cycle-6 auditor validated on 19 of 19 gates with three named cosmetic deviations. The replay fix is a foundational infrastructure unblock: from cycle 7 onward every sf2 replay proof across every song and instrument will correctly honour `profile.identity.program`.

1. Cycle 4 — family verdict, second profile, CRITICAL surfacing

1.1 The verdict

Cycle 4 opens as the adjudication cycle promised at cycle-3 close. The stage-2b leaderboard was already complete on disk (216 rows, 215 distinct render SHAs, 36 program-33 rows). Cycle 4 wrote `family_verdict_cg_bass.py`, ran it against the leaderboard, and emitted `data/v4/profiles/31a164f845f8e27e/bass_family_verdict.json`.

The pre-registered decision clauses evaluated as follows:

clause	value
top-1 embedding cosine ≥ 0.60	false (0.4946)
top-1 embedding cosine < 0.40	false
program 33 in top-3 by embedding cosine	false (rank 64)
program 33 not in top-5 by embedding cosine	true
top-1 preset in bass family {32..39}	false (program 19, organ)
spread ratio ≥ 0.10 vs cycle-1 coarse	true (1.75 $\times$)

Neither CONFIRMED nor RULED_OUT fired. The verdict is STILL_INDETERMINATE, emitted honestly at first-run under the frozen 0.60 / 0.40 thresholds.

1.2 The second pinned profile

Because the composite metric and the embedding metric name different presets, cycle 4 pinned a second profile alongside the cycle-2 organ pin. `bass_v2.json` carries `profile_id d62cd3b6-4521-5d4f-b840-87ef7800c48d`, program 33 Electric Bass Finger, gain 0.5, reverb 0.3, `post=EQ_only`, serialised at SHA `2a1cb340bfffd11016c566467b0d313fb...`. This is the top-1 by composite and the operator-facing “closest bass-shaped candidate” from the SoundFont family.

`bass_v2.replay_proof.json` was written under the cycle-6-canonical 7-key env-pin `env_pin_sha256 = 2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842a922ca` (the same 7-key subset used by cycle 2; the stage-2b sweep-time 9-key pin from cycle 3, which added `pyloudnorm_available` and `lufs_target_db`, is not consumed by the replay operation and does not enter the replay-proof hash). Verdict: `REPLAY_PROOF_HOLDS`, `run1 == run2 == 832868d0ea8a81cab2569e60445f80d516d1b5bb958b1b8b0c2e996bdb3aeac5`.

1.3 The CRITICAL defect

bass_v2.replay_proof.json records `run1 = run2 = 832868d0...3aeac5`. `bass.replay_proof.json` from cycle 2 recorded `832868d0...3aeac5` as well — the same SHA, on the same MIDI file, from two profiles that differ only in `identity.program` (17 vs 33). The verdict JSON records the finding under `canonical_replay_sha_equals_c2_note` as what the cycle-4 worker at that moment believed to be a benign program-invariance property: “`replay._replay_sf2` reads program from MIDI file’s own `program_change` events, not from the profile dict... `c1` rewrote `bass.mid` to embed `prog=33 program_change`, so any `sf2` profile invoked against `bass.mid` produces byte-identical audio.”

That framing was over-charitable. The property is not benign: it means the replay dispatch cannot distinguish two SoundFont profiles that intentionally pin different programs, which invalidates every SoundFont replay proof across every future song and instrument. The cycle-4 auditor upgraded the finding to a CRITICAL replay-defect chain (with cycle-2 as origin), and the finding entered cycle 5 as a manager escalation.

1.4 A second observation about environment pins

Cycle 4 also confirmed that the 7-key replay-time env pin (`PYTHONHASHSEED`, `SOURCE_DATE_EPOCH`, `TZ`, `LC_ALL`, single-thread `OMP / MKL / OPENBLAS`) is stable across the cycle-2 → cycle-3 transition even though the cycle-3 sweep-time env pin changed. This is a positive property of the replay-proof surface: the loudness library and target that entered the sweep are not consumed by the replay operation, so the replay-proof hash does not need to track them. Cycle 6 later ratifies this scoping choice as `_plan/env-pin-schema-unified-c6`.

2. Cycle 5 — the alternate family spike and the manager escalation

Cycle 5 was set up as a two-item cycle: put the CRITICAL defect on a formal escalation track, and stand up the alternate render family so cycle 6 could run both closures back-to-back.

Manager escalation. A `_manager` event was landed carrying the replay-invariance defect with CRITICAL severity, `action_required`, and a pre-registered fix contract for cycle 6 to close: replace the `_ = setup # documentation` line at `scripts/sound_match/replay.py` line 85 with a real MIDI-rewrite step; produce a regression test matrix with negative-inversion, positive-determinism, and existing- MIDI-neutrality cases (A/B/C); refresh both `sf2` replay proofs; and prove that the cross-profile SHAs now differ.

Family-2 spike. `scripts/sound_match/family2_stem_sampled_spike.py` was written (SHA `000c3ef68042f2da...6329e80`) as the shape probe for the stem-sampled render family. The spec fixed by the spike is:

- One slice of the reference bass stem is taken as the source timbre. The slice is centred on the `pyin`-estimated fundamental of the stem.
- The slice is pitch-shifted per note using a single-slice strategy — no per-note pick, no windowed `f0` refinement.
- A lightweight ADSR envelope is applied per note.
- The rendered take is loudness-normalised to `-18 LUFS-I`.

The spike also recorded a honest first observation about the reference stem: `pyin` estimated the fundamental at 41.20 Hz — the lower `fmin` bound, meaning the estimate is pinned to the floor rather than a true content estimate. This is a subharmonic latch; the spike declined to work around it and instead recorded it as an honest input property for the family verdict to inherit.

3. Cycle 6 — the two-track closure

3.1 Track 1 — replay-program-invariance fix

replay.py was patched in place at lines 79-93. The pre-patch line `_ = setup # documentation —` which discarded the program-select payload silently — was replaced by:

```
import mido # local import: only needed on sf2 dispatch
_mid = mido.MidiFile(str(midi_path))
for _tr in _mid.tracks:
    _to_del = [i for i, m in enumerate(_tr) if m.type == "program_change"]
    for i in reversed(_to_del):
        del _tr[i]
for _tr in _mid.tracks:
    if any(m.type == "note_on" for m in _tr):
        _tr.insert(0, mido.Message("program_change", channel=0,
                                   program=program, time=0))
        break
_rewritten = out_wav_path.with_suffix(".prog_forced.mid")
_mid.save(str(_rewritten))
cmd.append(str(_rewritten))
```

The rewrite is a deterministic pure function of (midi_path, program). It runs in-memory, saves the rewritten MIDI to a sibling .prog_forced.mid file next to the target WAV so the rewrite is auditable, and hands fluidsynth the rewritten file instead of the original.

File SHAs. replay.py moves from e03dad60d67fcf2fb93b5d01e026cc1d1b21a5496c27a110a0353fd5abf1fbab to 419d9558747eec61e58b3450b9f57b9bd057a7f8d7a31dfd1ab02f4d63c9f545.

Regression matrix. replay_fix_test_matrix.json records three tests, all passing:

- **A — negative inversion.** Same MIDI, two profiles with different identity.program values → different audio SHAs. Would have failed pre-fix.
- **B — positive determinism.** Same MIDI + same profile → same audio SHA on repeat, byte-identical.
- **C — existing-MIDI neutrality.** Rewriting a MIDI whose embedded program_change already matches the profile's identity.program yields byte-identical output before and after rewrite, so neutral cases do not shift.

Replay-proof refresh. Both cycle-2 and cycle-4 replay proofs were re-emitted under the post-fix replay.py. Pre-fix archives were preserved under pre_c6_fix/:

profile	pre-fix SHA	post-fix SHA	change
bass (program 17 organ, cycle 2 pin)	832868d0...3aeac5	c69775040c325b865be02931061b239f6238c877bae3f1b74ff0	CHANGED — fix took effect
bass_v2 (program 33 electric bass, cycle 4 pin)	832868d0...3aeac5	832868d0ea8a81cab2569e60415c4e4b5bb958b1b8b0c2e996bdb3ae	unchanged (expected — bass.mid already embeds program_change 33)

The cross-proof-differ property (bass ≠ bass_v2) is the audio- bytes evidence, independent of the test suite, that the fix reaches the render output.

Verdict. replay_fix_verdict.json records REPLAY_FIX_LANDS, three-way rubric-hash chain a9497ed585f9a8807ec0addb2d695b8b411eb60418c7bc3946663ce71c4178ef, all three regression tests pass, cross-proof SHAs differ, env-pin 2ac444c3...a922ca.

3.2 Track 2 — family-2 stem-sampled builder, profile, verdict

The cycle-5 spike was promoted to `scripts/sound_match/family2_stem_sampled_builder.py` and `scripts/sound_match/replay_family2.py`. Both files carry the project’s standard interpreter guard, no PRNG, and — per the sequential-only campaign discipline — no `--verify-det` flag.

The pinned family-2 profile data/v4/profiles/31a164f845f8e27e/bass_family2_v1.json:

- `profile_id`: 1f3c104a-2cc4-5e25-a802-d1360f1336ee (UUID5, deterministic function of the profile’s canonicalised parameter block).
- `render_family`: `stem_sampled_v1`.
- Lever set (frozen per the cycle-5 spec): `single_slice_pitch_shift`
 - `adsr_lite` + LUFS-I -18.
- `render_sha256_canonical_replay`: 9b4647cef61fe9d6...523276.

The replay proof holds byte-identically $\times 2$ at 9b4647cef61fe9d6...523276, under the same canonical 7-key env-pin 2ac444c3...a922ca.

The family verdict at data/v4/profiles/31a164f845f8e27e/bass_family2_verdict.json:

- `render_family`: `stem_sampled_v1`.
- `panel.mel_l1_db`: 7.6887.
- `panel.spectral_centroid_rmse_hz`: 3262.46.
- `panel.embedding_cos_vggish`: **0.0896**.
- `comparison_vs_sf2.sf2_top1_embedding_cos`: 0.4946.
- `comparison_vs_sf2.delta`: **-0.405**.
- `verdict`: **FAMILY2 RULED_OUT**.
- `rubric_hash`: 2dddc32a52d6fa4544a7153b0ac2709609180f0645f6e422a6ea72a2c7b91dfe.

The verdict fires cleanly under FD-1 first-run honesty: 0.0896 is 0.31 units below the 0.40 RULED_OUT threshold. Even if the subharmonic-latched pyin f0 were refined to a true fundamental, the single-slice-pitch-shift lever set is unlikely to lift the score by 0.31. The RULED_OUT verdict does not need the f0 estimate to be fixed first — it fires on the panel numbers as measured.

Family 2 is not universally invalid — the finding is per-cell. Future family-2 attempts on other songs and instruments remain in scope; on Chicken Grease bass, this specific lever set is empirically ruled out.

3.3 The env-pin unification

Every proof and verdict emitted in cycle 6 stamps the single canonical replay-time env pin `env_pin_sha256` = 2ac444c36298d6ada0579aba1a9160a5881703a4e628f5cccdd828b842a922ca. That covers the two refreshed sf2 replay proofs, the family-2 replay proof, the family-2 verdict, and the replay-fix verdict. This closes on the replay surface the “environment-pin scoping” observation raised by cycle 4. The cycle-3 sweep-time 9-key superset (`pyloudnorm_available` + `lufs_target_db`) is preserved as a diagnostic manifest under `_plan/env-pin-schema-unified-c6` rather than merged into the replay hash, because replay does not consume those two variables.

3.4 Audit acceptance

Cycle-6 auditor VALIDATED on 19 of 19 gates. Three deviations were named explicitly in the worker output and accepted as cosmetic:

- **Gate 8** — the manager supersede was carried through a two-step `action_required` → `in-progress` → `validated` transition path, with the supersede intent recorded on the `supersedes_`

path field of the validated event, because the ledger state-machine forbids the direct `action_required` → superseded transition. Substantive supersede semantics honoured.

- **Gate 17** — 14 cycle-6 ledger events landed rather than the brief’s 12; the extra event is the state-machine step required under gate 8. Substantive content count matches the brief.
- **Gate 19** — env-pin unification is scoped to the replay-proof surface, with the sweep-time superset preserved separately, rather than merged into a single hash. This is the operational closure the CRITICAL required.

Anchor preservation held: `bass.json`, `bass_v2.json`, `bass_family_verdict.json`, and the reference stem `data/v3/deliveries/31a164f845f8e27e/cert_run1/stems_6s/bass.wav` were all byte-identical pre and post cycle 6, along with the cycle-5 spike script. Storage delta $\approx$ 350 KB against the 500 MB per-cycle working-audio budget (0.07% utilisation).

4. State of Chicken Grease bass at cycle-6 close

Both frozen render families have been exercised:

- **SoundFont family:** `STILL_INDETERMINATE` at cycle-4 verdict (READ-ONLY thereafter). Top-1 by composite is program 33 Electric Bass Finger (`bass_v2.json`, `embedding_cos_vggish` = 0.4946 when measured against the program-19 top of the embedding metric — see §1). Top-1 embedding is program 19 Church Organ. Neither reaches the 0.60 CONFIRMED threshold.
- **Stem-sampled family v1** (`single_slice_pitch_shift` + `adsr_lite` + LUFFS-I –18): `FAMILY2_RULED_OUT` at cycle 6, `embedding_cos_vggish` = 0.0896.

No CONFIRMED profile exists. The M-V4-SHOWCASE-1 milestone is therefore blocked on cg-bass. Under the campaign’s operator- authority rule (FD-6, operator ear is the LANDS authority), the correct next step is a manager escalation to the operator with three named options: (1) accept the sf2 `STILL_INDETERMINATE` top-1 as the pinned cg-bass profile for showcase, unblocking M-V4-SHOWCASE-1; (2) refuse showcase until a further family lands, requiring either a different family-2 lever set (per-note pick or windowed f0 from the cycle-5 spec) or a new render family entirely (e.g. sample-based commercial VST via DawDreamer with sfizz fallback); (3) override the pre-registered 0.60 threshold, which is destructive to FD-1 pre-registration and requires an explicit operator override.

5. The replay fix as infrastructure unblock

The scope of the replay-program-invariance fix extends well beyond Chicken Grease bass. Every SoundFont-family replay proof for every song and instrument to come — the remaining four focus songs times six instruments per song, twenty-four profile cells with drums first per the operator’s instrument order — will run through the patched `replay.py`. Pre-fix, `_ = setup` # documentation would have produced silent program-invariance collisions on every one of them, exactly as it did between `bass.json` and `bass_v2.json`. The fix converts the SoundFont render family from “cannot ship replay proofs” to “can ship replay proofs on every song and instrument.” That is worth naming as the single largest infrastructure gain of these three cycles.

6. Open questions carried into cycle 7

1. **Cg-bass arc close-out.** Emit `data/v4/profiles/31a164f845f8e27e/bass_arc_closeout.json` recording both family verdicts (with their SHAs pinned), the best-available profile by any objective (`bass_v2.json` program 33, sf2 `embedding_cos` = 0.4946 — still below CONFIRMED), and the named blocker: no family CONFIRMED at frozen threshold;

operator policy call required.

2. **Manager escalation on M-V4-SHOWCASE-1 acceptance policy.** Emit `_manager/M-V4-SHOWCASE-1-cg-bass-acceptance-policy` with `status = action_required`, `severity = HIGH` and the three options listed in §4. Do not pick a default; operator picks per FD-6.
3. **Contingent advance.** If cycle-7 `live_guidance` carries an operator directive on option (1), open `M-V4-PROFILES-1/cg-drums-sweep-launched` as the next profile cell per the v4 spec instrument order (drums first after bass). Otherwise hold for operator input.

7. Two minor items noted but not acted on

- The pyin f0 estimate of 41.20 Hz on the reference bass stem is latched to the algorithm's `fmin` bound and therefore a subharmonic misidentification. The family-2 verdict fires regardless (see §3.2), and re-running the family-2 lever set with a corrected f0 would need to lift `embedding_cos_vggish` by 0.31 units — larger than the single-slice-pitch-shift lever set is likely to move. Recording as a first-class negative input property, not a defect to retry.
- The sweep-time 9-key env-pin superset lives under a diagnostic manifest rather than the canonical env-pin schema at `scripts/v3_spine/v3_pipeline/env_pin.py`. If a future milestone makes LUFs-integrated normalisation a general render-path invariant, the schema alignment belongs in `M-V4-CLOSE-1`.

8. Conclusions

Cycles 4–6 land the two hardest technical closures of the profiling milestone in a compact three-cycle sequence: the sf2 replay-program- invariance CRITICAL is fixed with cross-proof-differ evidence at the audio-bytes boundary, and the stem-sampled render family is empirically ruled out for Chicken Grease bass under the cycle-5 lever set. Both frozen families are now exhausted for this one cell, and the milestone reaches a defensible waypoint at which the operator's ear becomes the LANDS authority. Cycle 7 opens as a close-out plus operator-escalation cycle.

Appendix: implementation details

A.1 Files created or extended (cycles 4–6)

- `scripts/sound_match/emit_bass_v2_and_replay_proof.py` (cycle 4) — emits `bass_v2.json` + replay proof.
- `scripts/sound_match/family_verdict_cg_bass.py` (cycle 4) — runs the pre-registered decision protocol against the stage-2b leaderboard and writes `bass_family_verdict.json`.
- `scripts/sound_match/family2_stem_sampled_spike.py` (cycle 5) — shape probe, SHA 000c3ef68042f2da...6329e80, byte-identical pre==post cycle 6.
- `scripts/sound_match/family2_stem_sampled_builder.py` (cycle 6) — shipped stem-sampled builder promoting the cycle-5 spike.
- `scripts/sound_match/replay_family2.py` (cycle 6) — family-2 replay dispatch, sibling to `replay.py`.
- `scripts/sound_match/replay.py` (cycle 6, patched in place at L79–93) — moves from e03dad60... to 419d9558747eec61e58b3450b9f57b9bd057a7f8d7a31dfd1ab02f4d63c9f545.

A.2 Data artefacts

- `data/v4/profiles/31a164f845f8e27e/bass_family_verdict.json` (cycle 4) — STILL_INDETERMINATE; leaderboard stats and cross-metric top-3 tables embedded.

- data/v4/profiles/31a164f845f8e27e/bass_v2.json (cycle 4) — profile_id d62cd3b6-4521-5d4f-b840-87ef7800c48d, program 33, gain 0.5, reverb 0.3, post=EQ_only, profile SHA 2a1cb340bffd11016c566467b0d313fb002c5949ce881968702846867e090462.
- data/v4/profiles/31a164f845f8e27e/bass_v2.replay_proof.json (cycle 4, refreshed cycle 6) — run1 = run2 = 832868d0ea8a81cab2569e60445f80d516d1b5bb958b1b8b0c2e996bdb3aeac5; unchanged after fix, expected because bass.mid already embeds program_change 33.
- data/v4/profiles/31a164f845f8e27e/bass.replay_proof.json (refreshed cycle 6) — run1 = run2 = c69775040c325b865be029316d5ccbaaff6b3d2393b238c877bae3f1b74ff019c; changed from pre-fix 832868d0...3aeac5, definitive audio-bytes evidence the fix took effect.
- data/v4/profiles/31a164f845f8e27e/replay_fix_verdict.json (cycle 6) — REPLAY_FIX_LANDS, A/B/C = pass/pass/pass, cross-proof-differ = true, rubric hash a9497ed585f9a8807ec0addb2d695b8b411eb60418.
- data/v4/profiles/31a164f845f8e27e/replay_fix_test_matrix.json (cycle 6) — three-test regression matrix.
- data/v4/profiles/31a164f845f8e27e/bass_family2_v1.json (cycle 6) — profile_id 1f3c104a-2cc4-5e25-a802-d1360f1336ee, render_family = stem_sampled_v1, canonical replay SHA 9b4647cef61fe9d6...523276.
- data/v4/profiles/31a164f845f8e27e/bass_family2_v1.replay_proof.json (cycle 6) — run1 = run2 = 9b4647cef61fe9d6...523276.
- data/v4/profiles/31a164f845f8e27e/bass_family2_verdict.json (cycle 6) — FAMILY2_RULED_OUT, embedding_cos_vggish = 0.0896, delta_vs_sf2_top1 = -0.405, rubric hash 2dddc32a...91dfe.
- data/v4/profiles/31a164f845f8e27e/anchor_preservation_pre_c6.json and anchor_preservation_post_c6.json — all_match = true, n_mismatch = 0 on four READ-ONLY anchors + cycle-5 spike script SHA.
- data/v4/profiles/31a164f845f8e27e/pre_c6_fix/ — archived copies of pre-fix bass.replay_proof.json (932 B) and bass_v2.replay_proof.json (892 B).
- data/v4/profiles/31a164f845f8e27e/replay_c6_post_fix_sha.txt — post-fix replay.py SHA fingerprint.
- data/v4/profiles/31a164f845f8e27e/replay_fix_c6_rubric_hash.txt and family2_builder_c6_rubric_hash.txt — three-way rubric-hash chain anchors.

A.3 Environment pins in play across the arc

pin	scope
2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccc08bba2b72da	cycle-4 replay-time 7-key env-pin; stamped across every cycle-6 proof and verdict; closes cycle-4 replay-side pin drift on the replay surface.
d606c8bc0ebfd38bf64ce588e2b133f4a954556d3c5c92d257c3e53257faa2b	cycle-5 stage-2b sweep-time 9-key env-pin (adds pyloudnorm_available + lufs_target_db); preserved as diagnostic manifest under _plan/env-pin-schema-unified-c6, not merged into replay hash.

A.4 Fixed anchors preserved read-only across cycles 4-6

- Cycle-2 pinned bass profile bass.json (56cdc50a-..., program 17).
- Cycle-4 pinned bass profile bass_v2.json (d62cd3b6-..., program 33).
- Cycle-4 family verdict bass_family_verdict.json.
- Cycle-5 spike script family2_stem_sampled_spike.py (000c3ef68042f2da...6329e80).
- Reference bass stem WAV data/v3/deliveries/31a164f845f8e27e/cert_run1/stems_6s/bass.wav.

- scripts/palette_render/render_stem.py (214372d920a319a9...5b2b).
- Chicken Grease Method A operator-blessed WAV (cc919559b4508b6bfe86...).

A.5 Session references

Cycle 4: researcher 16c9463b-6761-4aa6-87cd-36fca4ec9109, worker da3b85eb-d4e5-4e57-ab6f-e9472a04c6f8, auditor 2fdae963-6c6a-4c5b-8e85-585bc3d444d8. Cycle 5: researcher 99dc41dc-2026-4f1a-a75e-78d41d57cc6d, worker 0feef7ba-2029-4964-8f65-1162c806cc1f, auditor 5ae30026-8675-41da-928d-a1194d595de2. Cycle 6: researcher c6145592-efdc-4068-9725-30150c7cf728, worker dd1a24f9-9b06-4b87-9003-c6a184521279, auditor 63e9e51f-4e28-4fad-bd1e-c75a975f79be.

A.6 Cross-reference map

Cycle-3 stage-2b leaderboard → cycle-4 family_verdict_cg_bass.py → bass_family_verdict.json (STILL_INDETERMINATE) → cycle-4 emit_bass_v2_and_replay_proof.py → bass_v2.json + first replay proof (SHA 832868d0..., colliding with cycle-2 replay proof) → cycle-4 auditor CRITICAL escalation → cycle-5 manager event + family2_stem_sampled_spike.py → cycle-6 Track 1 replay.py fix + regression matrix + refreshed proofs (bass moves to c69775040c..., bass_v2 stays at 832868d0..., cross-proof-differ) → replay_fix_verdict.json REPLAY_FIX_LANDS → cycle-6 Track 2 family2_stem_sampled_builder.py + replay_family2.py + bass_family2_v1.json + bass_family2_v1.replay_proof.json + bass_family2_verdict.json FAMILY2_RULED_OUT → cycle-7 close-out + operator escalation (open).